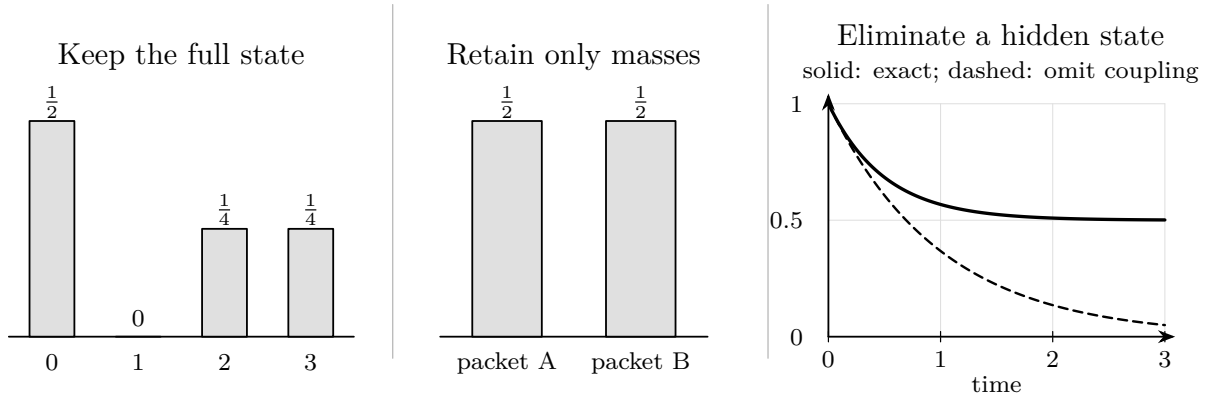
PAPER VI / A WORKED VISUAL ATLAS

Three ways to reduce a model

Main text: Sections 1–3, 8, and 13. Read the three operations separately.

This appendix follows small objects that can be computed completely. A four-state chain connects packet transfer, Markov aggregation, and Bayesian conditioning. A two-state coupled evolution exposes memory. A three-node Dirichlet problem makes the Green-operator normalization visible. These models are related by the paper's language, not by pretending that their state spaces are identical.



Left and center use one probability vector and its two block masses. The right panel is a separate retained/hidden model, developed on V – 22. Every curve is drawn from an explicit formula.

Coordinates retain; aggregation forgets; elimination remembers

Keeping every component $P_a x$ is an exact change of representation when $\sum_a P_a = I$. Reconstruct by summing the components. In contrast, replacing (p_0, p_1, p_2, p_3) by $(p_0 + p_1, p_2 + p_3)$ forgets how mass was distributed inside either pair. That information can matter for later observations.

Eliminating a hidden state is different again. Its coordinates disappear from the displayed state, but the exact retained equation acquires a history integral. Simply deleting the coupling produces the dashed curve above, not the original evolution.

Question	Worked route	Pages
When is coarse evolution exact?	Aggregate columns, then test intertwining.	V – 4–V – 6
When is a coarse posterior exact?	Compare two hidden shapes with the same masses.	V – 9–V – 13
What changes under a fractional power?	Keep modes, change spectral rates.	V – 15–V – 17
What returns after elimination?	Compute the self-energy and memory kernel.	V – 21–V – 25

Read this as: The paper supplies exact identities and conditional bounds. Whether a particular coarse model satisfies their hypotheses is a separate calculation.

READ BEFORE MULTIPLYING / CONVENTIONS

A dictionary with orientations

Main text: Sections 2, 4, 6, 7, and 12. Column probabilities and backward observables.

An apparent disagreement between two transfer formulas often comes from acting on different objects. Throughout this appendix, probabilities are columns and observables are functions written as columns. Their pairing is $f^{\mathsf{T}}p$. The transpose distinguishes the forward action on distributions from the backward action on observations.

Object	Exact meaning	What to check
p	A nonnegative column with $\mathbf{1}^{\mathsf{T}}p = 1$.	It is not a vector of function values.
Q	Continuous Markov generator.	Off-diagonal entries are nonnegative and columns sum to zero.
K	One discrete stochastic step.	$\mathbf{1}^{\mathsf{T}}K = \mathbf{1}^{\mathsf{T}}$ and $K_{ij} \geq 0$.
$K^{\mathsf{T}}f$	Expected next observable given the current state.	The same dynamics, with the opposite action on the dual object.
P_a	A coordinate packet projector.	All packets together reconstruct the full state.
C	Sums fine probability masses into coarse blocks.	Usually rectangular and not invertible.
E_π	Conditional expectation of observables.	A weighted average, self-adjoint for the π inner product.
$\mathcal{B}_\lambda$	$\mathrm{diag}(\lambda)p/(\lambda^{\mathsf{T}}p)$.	Nonnegative likelihood and positive total evidence.
L	A nonnegative self-adjoint operator.	Its evolution is e^{-tL}, not e^{tL}.
$R(z)$	$(zI - A)^{-1}$.	z belongs to the resolvent set.
$\mathcal{K}(t)$	The memory kernel after hidden-state elimination.	It is not the transition matrix K.

One convention test

For

$$K_2 = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \qquad p = e_0, \qquad f = (2,1)^{\mathsf{T}},$$

we obtain $K_2 p = (3/4, 1/4)^{\mathsf{T}}$ and

$$f^{\mathsf{T}}K_2 p = 7/4 = (K_2^{\mathsf{T}}f)_0.$$

With departure weights $w_i = e^{-V(i)}$, the backward Feynman–Kac operator is $\mathrm{diag}(w)K_2^{\mathsf{T}}$. Its forward dual is $K_2\,\mathrm{diag}(w)$. Interchanging these factors changes when the weight is applied.

> **Read this as:** Always name the object being transported. The transpose and the position of the likelihood or potential are mathematical choices, not typographic details.

CONTINUOUS TRANSFER / THE ENTIRE SOLUTION

A clock generated by one matrix

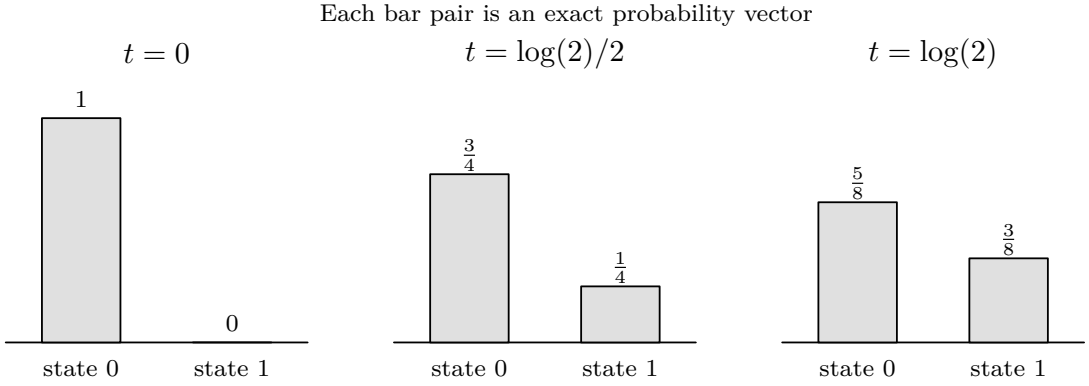
Main text: Theorems 1.2 and 4.1. Finite matrix exponentials and Markov positivity.

Take the generator

$$G = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The total-mass direction $(1, 1)$ has eigenvalue zero. The contrast $(1, -1)$ has eigenvalue -2. Resolving those two directions gives the full transfer, not only its first-order approximation:

$$T(t) = e^{tG} = \frac{1}{2}\begin{pmatrix} 1 + e^{-2t} & 1 - e^{-2t} \\ 1 - e^{-2t} & 1 + e^{-2t} \end{pmatrix}.$$



A point seed at state zero approaches the uniform distribution. The displayed times were selected so that every bar label is rational. This is continuous-time evolution, not a simulated trajectory.

Two checks explain the mechanism

The constant mode is unchanged, while each elapsed interval multiplies the contrast by e^{-2t}. Consequently two intervals multiply their contrast factors:

$$e^{-2s}e^{-2t} = e^{-2(s+t)}, \qquad T(s)T(t) = T(s+t).$$

For $t \geq 0$, the entries are nonnegative and every column sums to one. Differentiation gives $T'(t) = GT(t)$. These three facts separately check the initial condition, the evolution equation, and preservation of probability.

At $t = \log 2$, the retained first coordinate is $5/8$. It would instead be $1/2$ under the isolated scalar equation $\dot{x} = -x$. The discrepancy is the mass that can leave state zero and return. On V – 22, we remove the second coordinate correctly and recover exactly the same $5/8$ using a memory integral.

> **Read this as:** The semigroup lives on the complete state. One diagonal entry of a semigroup need not itself satisfy a scalar semigroup law.

FOUR-STATE MODEL / COORDINATES AND MODES

One evolution, two packet choices

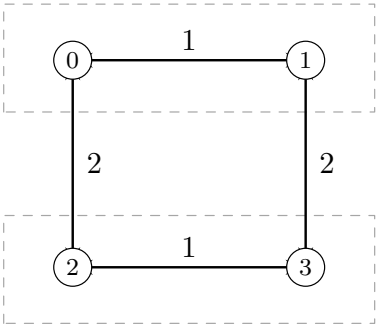
Main text: Theorem 2.1, Corollary 2.2, and the packet path expansion.

Our main finite model has generator

$$Q = \begin{pmatrix} -3 & 1 & 2 & 0 \\ 1 & -3 & 0 & 2 \\ 2 & 0 & -3 & 1 \\ 0 & 2 & 1 & -3 \end{pmatrix}.$$

The graph shows rates between actual states. The four normalized sign vectors on the right form an orthonormal basis u_j. Their projectors $P_j = u_j u_j^{\mathsf{T}}$ sum to the identity and satisfy $P_i P_j = 0$ for $i \neq j$.

Off-diagonal rates of Q



Top pair A; bottom pair B

Mode signs (each divided by 2)

	0	1	2	3	rate
constant	1	1	1	1	0
within	1	−1	1	−1	−2
between	1	1	−1	−1	−4
mixed	1	−1	−1	1	−6

State coordinates have off-diagonal transfers. Spectral packets reduce the generator and evolve independently. The two dashed state groups are the coarse partition used next.

The same evolution has a closed formula

Writing $q = e^{-2t}$ gives

$$e^{tQ} = P_0 + qP_1 + q^2 P_2 + q^3 P_3.$$

At $q = 1/2$, the point seed e_0 becomes $(15, 5, 9, 3)^{\mathsf{T}}/32$. The polynomial identity $T(q)T(r) = T(qr)$ checks composition exactly, without numerical matrix exponentiation.

Any other complete packet family still has exact block convolution. For the coordinate groups A and B,

$$P_b e^{(s+t)Q} P_a = \sum_{c \in \{\mathrm{A},\mathrm{B}\}} (P_b e^{sQ} P_c)(P_c e^{tQ} P_a).$$

The intermediate packet cannot generally be omitted. Only a reducing family eliminates all cross terms.

For later discrete calculations define $K = I + Q/8$. This matrix is stochastic, but *it is not* $e^{Q/8}$. It is the step kernel in the uniformization identity $e^{tQ} = e^{-8t} \sum_{n \geq 0} (8t)^n K^n / n!$.

> **Read this as:** Changing packet coordinates does not change the evolution. Replacing continuous time by one discrete step does.

EXACT LUMPABILITY / AGGREGATE COLUMNS

When two states are truly equivalent

Main text: Theorem 5.1. The hypothesis is an intertwining identity.

Group states $0, 1$ into A and states $2, 3$ into B. The aggregation map and the candidate coarse kernel are

$$C = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad \bar{K} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

A fine state in A has probability $1/4$ of entering B in one step, regardless of which fine state in A was occupied. The analogous statement holds for B.

$8K$

	0	1	2	3
0	5	1	2	0
1	1	5	0	2
2	2	0	5	1
3	0	2	1	5

CK: aggregate transition columns

	0	1	2	3
A	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
B	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$

$$\bar{K} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}$$

Fine trajectories differ. Coarse predictions agree.

Columns 0 and 1 agree; columns 2 and 3 agree.

The fine columns are visibly different. Their sums over each target packet agree within each source packet. That equality is the reason a two-state description works.

A one-step calculation controls every discrete horizon

The displayed columns give $CK = \bar{K}C$. Multiplication then yields

$$CK^n = \bar{K}^n C \qquad (n \geq 0).$$

For example, e_0 and e_1 have the same coarse initial vector $(1, 0)^\mathsf{T}$. Their fine evolutions differ, but their coarse vectors agree at every step. No choice of the within-A conditional distribution is needed for the prediction.

The continuous generator also closes:

$$CQ = \bar{Q}C, \qquad \bar{Q} = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}.$$

The Duhamel identity therefore gives $Ce^{tQ} = e^{t\bar{Q}}C$. This is a second consequence of the corresponding generator identity, not an inference that K is a fixed-time sample of Q.

Aggregation is still not invertible. The theorem says the discarded details are irrelevant for *these predictions*. It does not say they are irrelevant to every later likelihood. The observation counterexample on V – 10 uses this exact same partition.

> **Read this as:** A partition is not automatically a valid Markov model. The aggregate transition columns must be constant within each source block.

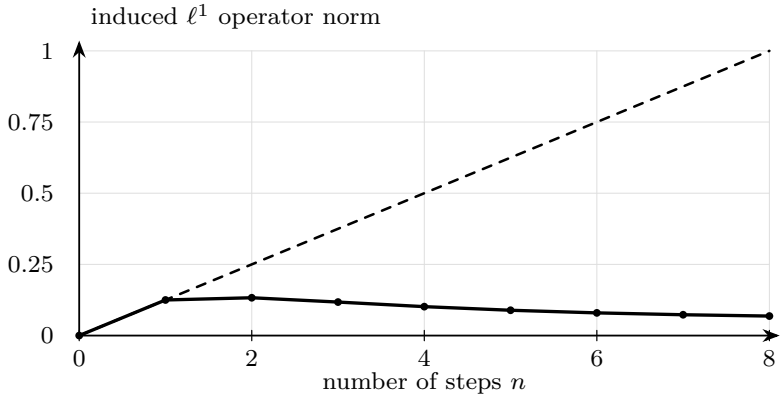
DEFECTS / A GUARANTEE WITHOUT A FIT

A broken lump and its error budget

Main text: Theorems 3.1, 3.3, and 5.2. Continuous and discrete defect identities.

Modify only column zero of K: move $1/16$ of its mass from target state zero to target state two. The perturbed matrix K_ε remains stochastic, but its aggregate column now differs from column one. Relative to the unchanged coarse kernel,

$$\Delta = CK_\varepsilon - \bar{K}C = \begin{pmatrix} -1/16 & 0 & 0 & 0 \\ 1/16 & 0 & 0 & 0 \end{pmatrix}, \qquad \|\Delta\|_1 = 1/8.$$



$\|\Delta\|_1 = 1/8$

solid: exact defect
dashed: $n/8$

The bound is a guarantee, not a fitted trend.

The solid values are computed with rational matrix powers. The dashed certificate uses only stochasticity and the exact one-step defect. It need not be sharp.

Telescoping tells us where every error originates

For $n \geq 0$,

$$CK_\varepsilon^n - \bar{K}^n C = \sum_{j=0}^{n-1} \bar{K}^{n-1-j} \Delta K_\varepsilon^j.$$

Each summand propagates one defect between an earlier fine evolution and a later coarse evolution. Column-stochastic matrices have induced ℓ^1 norm one, so the norm is at most $n/8$. The empty sum at $n = 0$ is zero.

The continuous counterpart is equally explicit. With fine generator -1, coarse generator -2, and $C = 1$, the generator defect is 1 and

$$e^{-t} - e^{-2t} = \int_0^t e^{-2(t-r)} e^{-r}\, dr.$$

Its resolvent version is $1/(z+1) - 1/(z+2) = 1/((z+1)(z+2))$ for $z > 0$.

> **Read this as:** A small defect does not disappear. The identity specifies how it accumulates, and the norm assumptions turn that identity into a finite-horizon certificate.

OBSERVABLES / WEIGHTED ORTHOGONALITY

Averaging is not summing

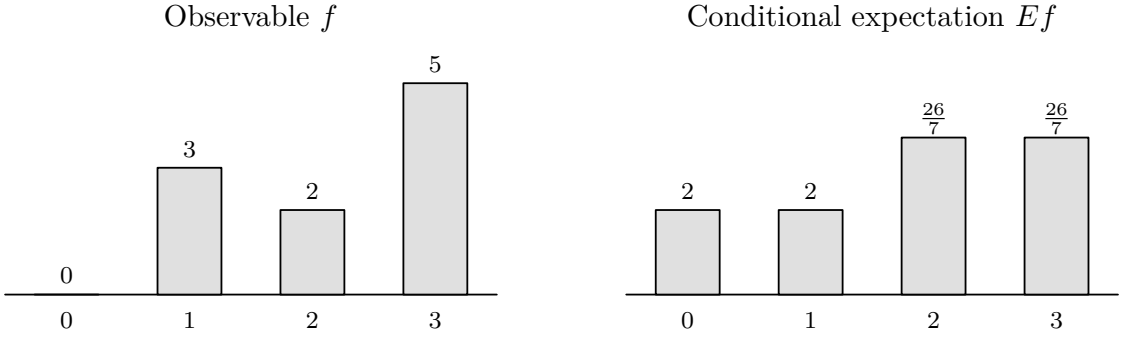
Main text: Theorem 6.1. Strictly positive weights are part of the setting.

For this observable example, choose the fixed probability weights $\pi = (1, 2, 3, 4)/10$. They are not being identified with the stationary law of the earlier symmetric chain. On the same two-block partition, conditional expectation is

$$E = \begin{pmatrix} 1/3 & 2/3 & 0 & 0 \\ 1/3 & 2/3 & 0 & 0 \\ 0 & 0 & 3/7 & 4/7 \\ 0 & 0 & 3/7 & 4/7 \end{pmatrix}.$$

It replaces function values by their *weighted block average*. In particular, $f = (0, 3, 2, 5)^{\mathsf{T}}$ becomes $Ef = (2, 2, 26/7, 26/7)^{\mathsf{T}}$.

Weights $\pi = (1, 2, 3, 4)/10$; the block values are 2 and 26/7.



The bars are function values, not state probabilities. Their total need not be one. The block masses 3/10 and 7/10 determine the denominators in the averaging formula.

Which inner product makes this an orthogonal projection?

Put $W = \mathrm{diag}(\pi)$. Direct multiplication checks

$$E^2 = E, \qquad WE = E^{\mathsf{T}}W, \qquad E\mathbf{1} = \mathbf{1}.$$

The middle equation is weighted self-adjointness. Ordinary symmetry fails: $E_{01} = 2/3$ whereas $E_{10} = 1/3$. Testing $E = E^{\mathsf{T}}$ would therefore reject a correct conditional-expectation projector.

The range consists exactly of functions constant on each block. The error $f - Ef$ has weighted mean zero inside every block, so it is weighted-orthogonal to every block-constant function. This is the finite projection interpretation of conditional expectation.

In contrast, C acts on probability columns by summing masses. The shapes of C and E differ, and their domains have different interpretations. The notation "coarsening" alone does not specify which operation is intended.

> **Read this as:** An exact weighted projector can be oblique in the unweighted Euclidean picture. Keep the measure and the inner product in the model.

MARTINGALE PACKETS / THREE EXACT PIECES

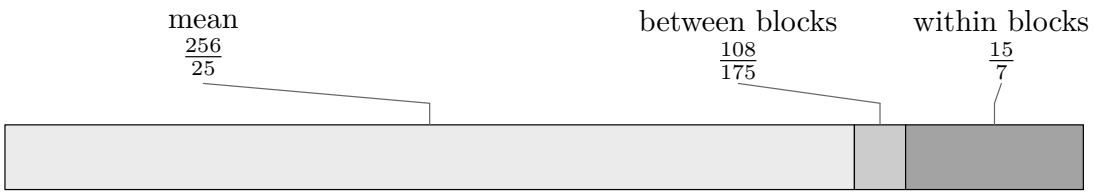
Where the variance goes

Main text: Theorem 6.2. Nested weighted projections and their differences.

Keep the weights and observable from V–7. Let $E_0 f$ be the constant global mean, let $E_1 = E$ average within A and B, and let $E_2 = I$ retain every value. Define

$$D_0 = E_0, \qquad D_1 = E_1 - E_0, \qquad D_2 = I - E_1.$$

Then $D_0 f$ is the mean, $D_1 f$ describes the contrast between block means, and $D_2 f$ is the fluctuation left inside each block.



$$D_0 f = \begin{bmatrix} \frac{16}{5} \\ \frac{16}{5} \\ \frac{16}{5} \\ \frac{16}{5} \end{bmatrix} \qquad D_1 f = \begin{bmatrix} -\frac{6}{5} \\ -\frac{6}{5} \\ \frac{18}{35} \\ \frac{18}{35} \end{bmatrix} \qquad D_2 f = \begin{bmatrix} -2 \\ 1 \\ -\frac{12}{7} \\ \frac{9}{7} \end{bmatrix}$$

The strip lengths are proportional to the three exact weighted energies. The between-block component is small but not zero. The vectors below specify the corresponding directions, not only their sizes.

The energy identity follows from actual orthogonality

The tower identities give $E_0 E_1 = E_1 E_0 = E_0$. Together with weighted self-adjointness, these imply

$$D_i D_j = 0 \ (i \neq j), \qquad \sum_{j=0}^{2} D_j = I.$$

For the chosen f, the total weighted energy is $\|f\|_\pi^2 = 13$ and the mean is $16/5$. Consequently

$$\mathrm{Var}_\pi(f) = 13 - \left(\frac{16}{5}\right)^2 = \frac{69}{25} = \frac{108}{175} + \frac{15}{7}.$$

The first term is the energy of the between-block contrast, and the second is the weighted within-block variance. This is a decomposition of one observable in one fixed weighted space.

Changing the probability weights between levels would change the inner products and the averaging operators. The displayed orthogonal decomposition would no longer follow merely from nesting the sets. Likewise, choosing unrelated projections can preserve the telescope $\sum D_j = I$ while losing orthogonality and the energy sum.

> **Read this as:** The differences between nested conditional expectations are not just bookkeeping. They are orthogonal, separately measurable parts of the original signal.

BAYESIAN TRANSFER / AN EXACT POSTERIOR

Evidence changes shape and mass

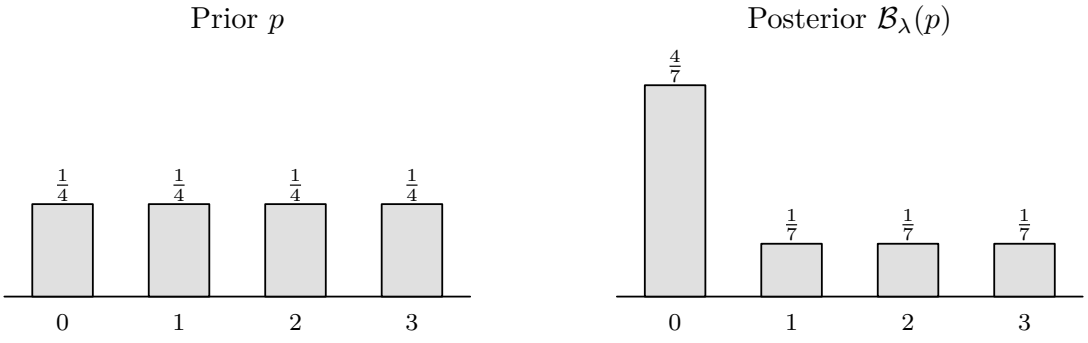
Main text: Theorems 7.1–7.3. Normalization, packet reconstruction, and positive evidence.

Start with $p = (1,1,1,1)^{\mathsf{T}}/4$ and likelihood weights $\lambda = (4,1,1,1)^{\mathsf{T}}$. The unnormalized vector is $(1,1/4,1/4,1/4)^{\mathsf{T}}$, its evidence is $Z = 7/4$, and

$$\mathcal{B}_\lambda(p) = \frac{(4,1,1,1)^{\mathsf{T}}}{7}.$$

Likelihood weights may be rescaled, so an evidence larger than one is not a probability claim. Multiplying all likelihoods by one common positive factor leaves this posterior unchanged.

$\lambda = (4,1,1,1)$; evidence $Z = 7/4$



The A packet becomes heavier and its internal shape changes. The B packet retains an equal internal split but loses total mass. These are different effects of the same likelihood.

Reconstruct from packet mass times packet shape

The unnormalized packet evidences are $Z_{\mathrm{A}} = 5/4$ and $Z_{\mathrm{B}} = 1/2$. The resulting masses and conditional shapes are

$$m_{\mathrm{A}} = 5/7, \quad q_{\mathrm{A}} = (4/5, 1/5, 0, 0)^{\mathsf{T}}, \qquad m_{\mathrm{B}} = 2/7, \quad q_{\mathrm{B}} = (0, 0, 1/2, 1/2)^{\mathsf{T}}.$$

Thus $m_{\mathrm{A}} q_{\mathrm{A}} + m_{\mathrm{B}} q_{\mathrm{B}}$ is exactly the displayed posterior. Keeping only the masses would not reconstruct the fine state.

There is no need to divide by a zero packet evidence. For likelihood $(0,0,1,1)$, packet A has mass zero and contributes the zero vector. Only the occupied B packet has a conditional probability vector. If the *total* evidence is zero, however, the Bayesian update is not defined and the implementation rejects the input.

> **Read this as:** Bayesian reconstruction uses two kinds of data: how much posterior mass a packet carries and how that mass is distributed inside the packet.

SUFFICIENCY / A COMPUTABLE FAILURE WITNESS

Two priors that fool the coarse model

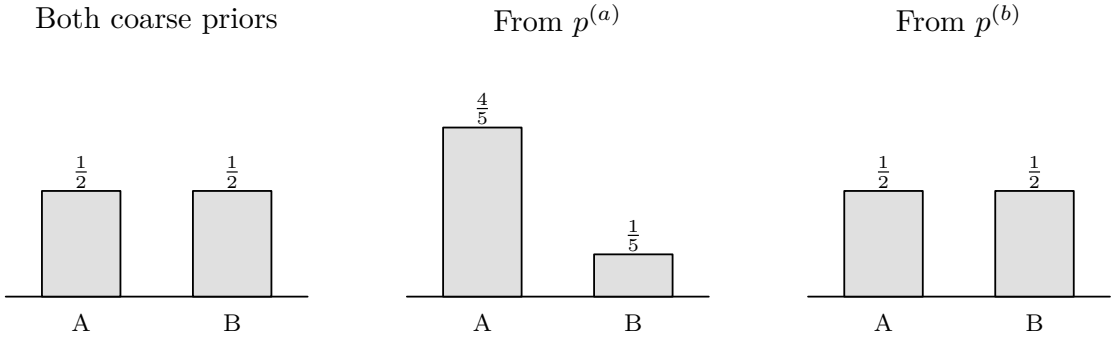
Main text: Theorem 8.2 and the one-packet exception.

Keep the same likelihood $\lambda = (4, 1, 1, 1)^{\mathsf{T}}$ and partition A= $\{0, 1\}$, B= $\{2, 3\}$. Compare

$$p^{(a)} = (1/2, 0, 1/2, 0)^{\mathsf{T}}, \qquad p^{(b)} = (0, 1/2, 1/2, 0)^{\mathsf{T}}.$$

Both have coarse prior $(1/2, 1/2)^{\mathsf{T}}$. A procedure given only that two-entry vector cannot distinguish them.

Hidden redistribution changes the packet-A posterior from 1/2 to 4/5.



After the identical fine likelihood is applied, the coarse posteriors are $(4/5, 1/5)$ and $(1/2, 1/2)$. A single deterministic update of the common coarse prior cannot produce both.

Why this witnesses the necessity of blockwise constancy

The A evidence for $p^{(a)}$ is 2, whereas it is 1/2 for $p^{(b)}$. Their B evidence is 1/2 in both cases. The posterior odds therefore distinguish the hidden distribution inside A.

For strictly positive fine and coarse likelihoods, and at least two coarse blocks, the manuscript proves the all-prior criterion

$$C\mathcal{B}_\lambda(p) = \mathcal{B}_{\bar{\lambda}}(Cp) \quad \text{for every } p \quad \Longleftrightarrow \quad \lambda_i = c\bar{\lambda}_a \quad (i \in I_a)$$

for one global $c > 0$. The global scale is invisible after normalization. A different arbitrary scale for each packet is not invisible: it changes posterior odds.

If all states are merged into a single packet, every normalized posterior coarsens to (1). That degenerate commutation holds for every likelihood and cannot identify any within-packet information. It is explicitly excluded from the nontrivial criterion.

> **Read this as:** A prior-dependent effective likelihood can always summarize the update. A fixed likelihood that works for every prior is a much stronger requirement.

BAYESIAN DEFECT / MORE THAN AN ERROR BAR

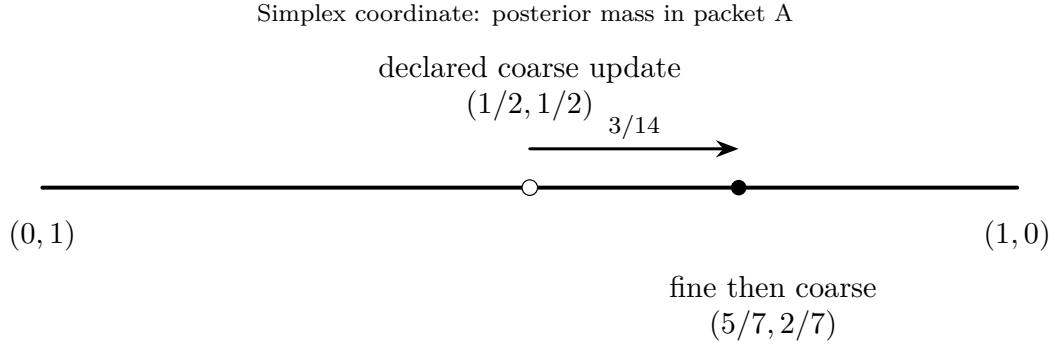
The exact information-loss vector

Main text: Theorems 8.1 and 8.4, and Corollary 8.6.

For a prior p, let $P = Cp$ and let $r_a(p)$ be the conditional average of the fine likelihood inside packet a. For the uniform prior and $\lambda = (4, 1, 1, 1)$,

$$P = (1/2, 1/2)^{\mathsf{T}}, \qquad r = (5/2, 1)^{\mathsf{T}}.$$

The exact coarse posterior is $\mathcal{B}_r(P) = (5/7, 2/7)^{\mathsf{T}}$. Suppose instead that the declared coarse likelihood is $\bar{\lambda} = (1, 1)$.



The posterior error lies along the mass-preserving direction of the two-state simplex. Its entries are opposite because both endpoints have total mass one.

Compute the factorization, including the null direction

Here $\bar{b} = \mathcal{B}_{\bar{\lambda}}(P) = P$, $\delta = r - \bar{\lambda} = (3/2, 0)^{\mathsf{T}}$, and $Z_r = 7/4$. Hence

$$\frac{1}{Z_r}\left(\operatorname{diag}(P) - \bar{b}P^{\mathsf{T}}\right)\delta = \frac{4}{7}\begin{pmatrix} 1/4 & -1/4 \\ -1/4 & 1/4 \end{pmatrix}\begin{pmatrix} 3/2 \\ 0 \end{pmatrix} = \begin{pmatrix} 3/14 \\ -3/14 \end{pmatrix}.$$

This is the actual fine-to-coarse discrepancy, not a fitted approximation. The matrix kills $\bar{\lambda}$, recording the common-scale null direction. Its ℓ^1 error is $3/7$.

For a family of likelihoods, states may be grouped by equality of their *whole likelihood profile*. The family $(3, 3, 1, 1)$ and $(1, 2, 1, 2)$ separates all four states even though either likelihood alone permits a coarser grouping. The common refinement is therefore the discrete partition. The theorem's one-packet exception still remains.

> **Read this as:** The defect factors through exactly the conditional evidence information missing from the declared coarse model. A new observation family can require a finer partition.

BAYESIAN FLOW / COMPOSITION WITHOUT LINEARITY

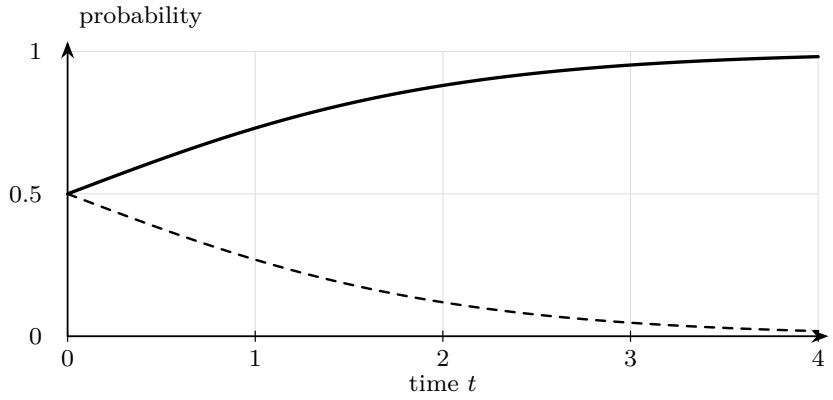
A nonlinear semigroup of odds

Main text: Theorems 7.1 and 9.1. Likelihood composition and the replicator field.

Let the score vector be $g = (1, 0)$ and start at $(1/2, 1/2)$. The likelihood at time t is $(e^t, 1)$, so the first posterior coordinate is

$$p_0(t) = \frac{e^t}{1 + e^t}, \qquad p_1(t) = \frac{1}{1 + e^t}.$$

The odds p_0/p_1 grow exponentially, while the probabilities remain inside the simplex. At $t = \log 2$ the posterior is exactly $(2/3, 1/3)$.



The solid and dashed curves sum to one. The curvature is caused by the shared evidence denominator, not by a nonlinear likelihood multiplication law.

Two properties that must not be conflated

If $a > 0$ multiplies the odds, the first-coordinate map is

$$\phi_a(x) = \frac{ax}{1 + (a - 1)x}.$$

Direct algebra gives $\phi_a(\phi_b(x)) = \phi_{ab}(x)$. Thus $a = e^t$ gives a nonlinear semigroup. Nevertheless, $\phi_3(1/2) = 3/4$, while the mean of $\phi_3(0)$ and $\phi_3(1)$ is $1/2$. Superposition fails.

Differentiation of the normalized flow yields the centered score equation

$$\dot{p}_i = (g_i - \langle g, p \rangle) p_i.$$

For the displayed model, $\dot{p}_0 = p_0(1 - p_0)$. The negative average-score term is what keeps the total derivative zero. Omitting the derivative of the normalizer would not preserve mass.

Likelihood composition also preserves the evidence identity: the first evidence times the second conditional evidence equals the combined evidence. Normalization cancels in the posterior, but that factor still matters when comparing data models.

> **Read this as:** A semigroup is a composition law in time. It does not imply that the maps are linear or that they preserve mixtures.

HIDDEN MARKOV FILTERING / TIME ORDER

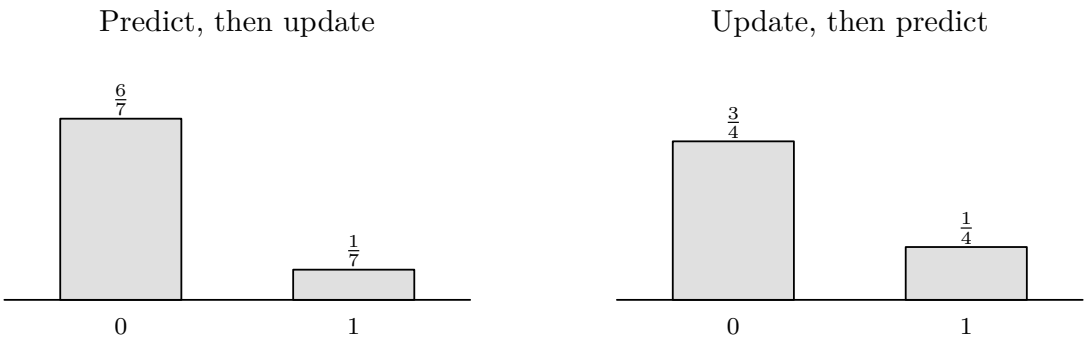
Prediction and observation do not commute

Main text: Proposition 9.3. Linear prediction followed by a positive-evidence update.

Use the two-state kernel K_2 from V – 2, the initial distribution e_0, and likelihood $(1, 1/2)$. Predicting first gives $(3/4, 1/4)$. Its likelihood-weighted vector is $(3/4, 1/8)$, with evidence $7/8$, so the posterior is $(6/7, 1/7)$.

Updating first leaves e_0 unchanged. Predicting afterward gives $(3/4, 1/4)$ instead. The order of operations expresses which time the observation concerns.

Same prior e_0, transition K_2, and likelihood $(1, 1/2)$



Both routes produce valid probability vectors. Their difference is therefore not a normalization bug. It is a different conditioning problem.

Keep an unnormalized path alongside the filter

For a sequence of observations, let $D_r = \operatorname{diag}(\lambda^{(r)})$. The forward recursion is

$$\widehat{p}_r = K p_{r-1}, \qquad p_r = D_r \widehat{p}_r / Z_r.$$

Its unnormalized companion is

$$v_r = D_r K \cdots D_1 K p_0, \qquad p_r = \frac{v_r}{\mathbf{1}^{\mathsf{T}} v_r}, \qquad \mathbf{1}^{\mathsf{T}} v_r = \prod_{j=1}^{r} Z_j.$$

The product of evidences records the normalization lost at each individual step. The four-state computation on V – 26 checks both paths independently, including their coarse versions.

Here $D_r K$ means an observation after prediction. In the Feynman–Kac example on V – 20, departure-state weighting has forward operator $K D_w$. Both conventions are legitimate once their meanings are fixed, but they cannot be substituted for one another.

> **Read this as:** Write the chronological sequence before multiplying matrices. A product can be normalized correctly and still answer the wrong time-ordered question.

BETA MODEL / POSTERIOR AND EVIDENCE

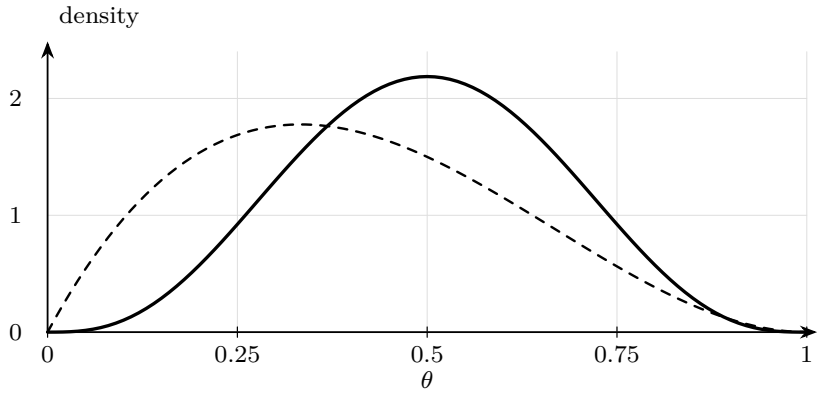
Conjugacy tracks exponents

Main text: Example 9.2. A finite-parameter conjugate family, not a universal posterior model.

For a small exact example, take the Beta$(2, 3)$ density

$$\rho(\theta) = 12\theta(1-\theta)^2, \qquad 0 \le \theta \le 1.$$

Observe a specified ordered sequence with two successes and one failure. Its likelihood is $\theta^2(1-\theta)$. Multiplication adds exponents, so the posterior belongs to Beta$(4, 4)$ with density $140\theta^3(1-\theta)^3$.



Prior: Beta$(2, 3)$

Posterior: Beta$(4, 4)$

ordered data: $3/35$
count data: $9/35$

Same posterior,
different evidence.

The dashed prior is asymmetric. The solid posterior is symmetric with mean $1/2$. Polynomial integration verifies the normalizers exactly; the curves are sampled drawings of those polynomials.

The observation convention changes the evidence

The integral of the unnormalized posterior is

$$\int_0^1 12\theta^3(1-\theta)^3\,d\theta = \frac{3}{35}.$$

Dividing by $3/35$ gives the displayed normalized density. If the observation is instead the *count* of two successes in three trials, there are $\binom{3}{2} = 3$ compatible ordered sequences. The count likelihood is therefore $3\theta^2(1-\theta)$ and its evidence is $9/35$.

The extra factor does not depend on θ, so it cancels from the posterior. The two experiments thus have the same posterior distribution but different data probabilities. Evidence is not determined by the final normalized density alone.

The exact check also integrates both densities to one and integrates θ against the posterior to obtain $1/2$. This demonstrates one conjugate family in elementary terms. It does not show that arbitrary likelihoods preserve that family or that all posterior information can be compressed into two exponents.

> **Read this as:** Conjugacy gives a closed parameter update when a specified likelihood respects a specified family. Preserve the evidence as well as the normalized posterior.

FRACTIONAL DIFFUSION / NEW COUPLINGS

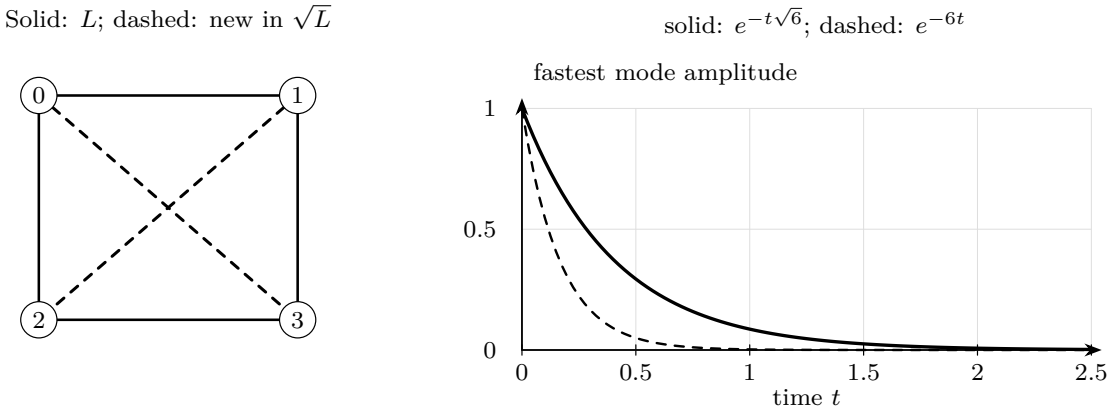
The modes stay; their clocks change

Main text: Theorem 10.1. A positive self-adjoint finite model and its spectral powers.

Return to the four-state generator on V − 4 and put $L = -Q$. It is real symmetric and nonnegative, with orthogonal projectors P_0, P_1, P_2, P_3 and eigenvalues $0, 2, 4, 6$. Its square root is

$$L^{1/2} = \sqrt{2}P_1 + 2P_2 + \sqrt{6}P_3, \qquad (L^{1/2})^2 = L.$$

Both heat evolutions retain these modes. Ordinary diffusion multiplies mode j by e^{-2jt}; square-root diffusion multiplies it by $e^{-\sqrt{2j}\,t}$. The constant mode is unchanged in both.



The highest mode decays at rates 6 and $\sqrt{6}$. The graph beside the curves shows the additional couplings in the fractional operator. Curves display the explicit formulas, not an empirical fit.

Fractional does not mean entrywise

Although $L_{03} = 0$, its spectral square root satisfies

$$(L^{1/2})_{03} = \frac{\sqrt{6} - \sqrt{2} - 2}{4} < 0.$$

Thus $-L^{1/2}$ permits a direct transition between states 0 and 3, which were not neighbors for Q. Taking the square root of each matrix entry would neither produce this value nor square back to L.

The exact block aggregation remains compatible in this example:

$$CL^{1/2} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} C.$$

The coarse matrix is the positive square root of $-Q_c$. This is a concrete naturality identity, checked on the full matrices. It is not a claim that an arbitrary partition commutes with every fractional construction.

The computation uses a nonnegative self-adjoint operator and a specified spectral branch. General nonnormal matrices, unbounded domains, and positivity of other fractional evolutions need their own hypotheses.

> **Read this as:** Fractional powers alter modal decay rates. Reassembling those modes can introduce long-range spatial interactions even when the original generator was sparse.

FRACTIONAL DEFECTS / NEAR ZERO VERSUS AWAY FROM ZERO

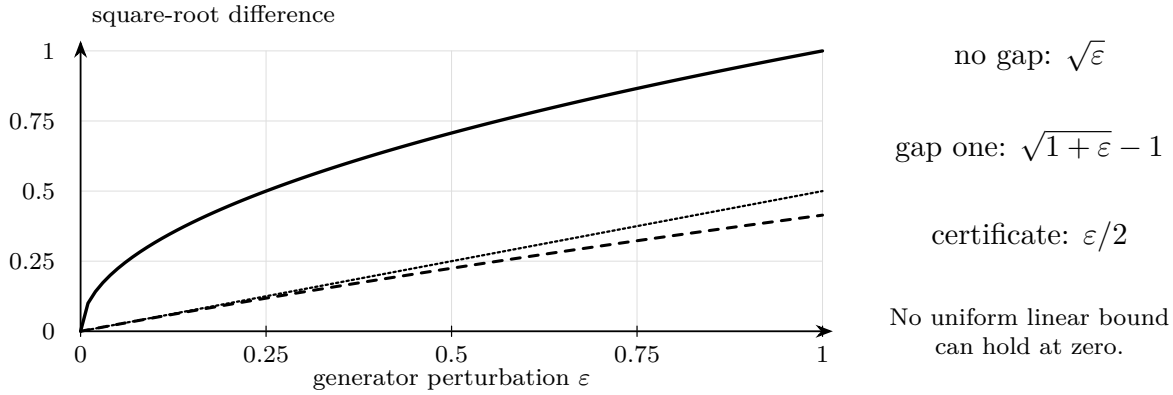
A gap changes the error scale

Main text: Theorem 10.3 and Corollary 10.4. Separate generator error from fractional error.

The function $a \mapsto \sqrt{a}$ changes most rapidly near zero. A scalar model already shows why a spectral gap appears in fractional-defect estimates:

$$|\sqrt{\varepsilon} - \sqrt{0}| = \sqrt{\varepsilon}, \qquad \sqrt{1+\varepsilon} - 1 = \frac{\varepsilon}{\sqrt{1+\varepsilon}+1} \leq \frac{\varepsilon}{2}.$$

The same input difference has a square-root scale at zero and a linear bound above a fixed positive lower bound.



The upper curve compares 0 with ε. The lower curve compares 1 with $1+\varepsilon$. The dotted line $\varepsilon/2$ bounds the latter. All plotted values come from the displayed scalar formulas.

Read an operator bound with its hypotheses

For positive self-adjoint finite matrices $A, B \geq mI$ and $0 < \alpha < 1$, the manuscript's gapped estimate has the form

$$\|CA^{\alpha} - B^{\alpha}C\| \leq \alpha m^{\alpha-1} \|CA - BC\|.$$

It concerns an intertwining defect, not just two eigenvalue lists. With $C = 1$, $A = 4$, $B = 9$, $m = 4$, and $\alpha = 1/2$, the generator defect has size 5, the fractional defect has size 1, and the bound is $5/4$.

The common zero mode of a Markov generator cannot simply be included in a positive-gap estimate. One must either work on an appropriate gapped subspace, handle the zero mode separately, or use the gapless estimate. In particular, a good estimate on each finite level need not have a constant uniform in the refinement level.

On V – 27, the sequences $1/n^2$ and $1/n$ turn this change in error scale into a concrete distinction between summable and nonsummable defects.

> **Read this as:** A fixed gap is quantitative information. It prevents the derivative of the fractional function from becoming singular along the comparison.

SUBORDINATION / A DISTRIBUTION OF ORDINARY TIMES

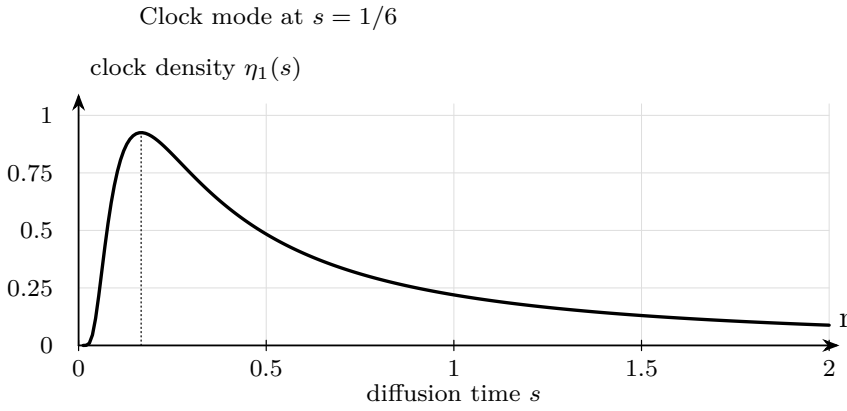
A random diffusion clock

Main text: Theorem 10.2. The Poisson subordination formula for the finite positive model.

Square-root evolution can be written as an average of ordinary diffusion over a positive clock variable s:

$$e^{-t\sqrt{L}} = \int_0^\infty \eta_t(s) e^{-sL}\, ds, \qquad \eta_t(s) = \frac{t}{2\sqrt{\pi}} s^{-3/2} e^{-t^2/(4s)}, \quad t > 0.$$

Here t is the external time; s is the ordinary diffusion time being mixed. They are not interchangeable variables.



$$\int_0^\infty \eta_t(s)\, ds = 1$$

$$\int_0^\infty s\eta_t(s)\, ds = \infty$$

A probability clock need not have a finite mean.

The density η_1 peaks at $s = 1/6$ but has a long right tail. The drawing ends at $s = 2$; it does not claim that the missing tail has zero mass.

Check the normalizer before interpreting a density

Under $s = t^2/(4u^2)$, reversing the integration limits gives

$$\eta_t(s)\,|ds| = \frac{2}{\sqrt{\pi}} e^{-u^2}\, du.$$

Its integral over $u > 0$ is one, so η_t is a probability density. On each eigenmode of L, the operator formula reduces to the scalar Laplace identity

$$\int_0^\infty \eta_t(s) e^{-\lambda s}\, ds = e^{-t\sqrt{\lambda}}.$$

The manuscript supplies that scalar identity and its spectral assembly. The appendix separately checks the change of variables and normalization; sampled curves are not used to certify an improper integral.

A normalized density need not have a finite mean. For large s, the exponential factor tends to one and $\eta_t(s)$ is proportional to $s^{-3/2}$. Consequently $s\eta_t(s)$ has a nonintegrable $s^{-1/2}$ tail. The clock has infinite mean even though its total mass is one.

> **Read this as:** Subordination is a weighted mixture of diffusion times. Replacing the clock distribution by a single average time is not justified, and here that average is infinite.

DIRICHLET GRID / THE MISSING FACTOR OF h

A Green kernel is an inverse with units

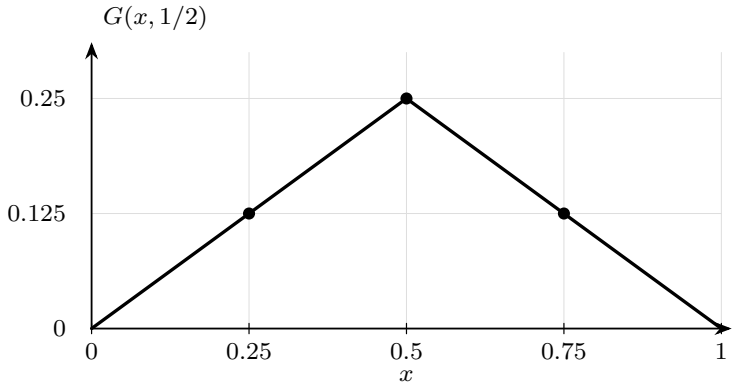
Main text: Theorem 11.1. Exact nodal Green reconstruction for this one-dimensional stencil.

Take three interior nodes $x_i = i/4$ on $[0, 1]$, with zero boundary values and $h = 1/4$. The discrete negative second derivative is

$$L_h = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

For the continuous Dirichlet problem, the Green function is $G(x, y) = \min(x, y) - xy$. At these nodes the exact comparison is

$$L_h^{-1} = h\,[G(x_i, x_j)]_{i,j=1}^{3} = \frac{1}{64} \begin{pmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix}.$$



$64L_h^{-1}$, $h = 1/4$

3	2	1
2	4	2
1	2	3

$L_h^{-1} = hG_{\text{nodes}}$

The tent is the central column of the continuous Green kernel. It has nodal values $(1, 2, 1)/8$; its discrete inverse column is $(1, 2, 1)/32$, smaller by the factor h.

A forcing vector is not a quadrature-weighted measure

For sampled forcing f_j, the finite solution is

$$u_i = \sum_j (L_h^{-1})_{ij} f_j = h \sum_j G(x_i, x_j) f_j.$$

The factor h converts samples into the quadrature weights appropriate to this formula. Dropping it multiplies the computed solution by four on the displayed grid.

A discrete unit coordinate vector and a discretized unit point mass use different normalizations. For $f = e_2$, the result is the second inverse column. For $f = e_2/h$, it is the unscaled tent sampled at the nodes. Either convention can be useful, but it must be specified before comparing answers.

Exact checks multiply L_h by the proposed inverse at $1, 3, 7$, and 15 interior nodes. The general identity is the manuscript theorem. These finite checks do not extend it to arbitrary meshes, dimensions, or boundary conditions.

> **Read this as:** An inverse kernel carries a measure convention. A correct-looking tent can give a wrong solution if the grid weight is omitted.

FINITE DIFFERENCES / A QUARTIC TEST PROBLEM

Convergence with a nonzero exact error

Main text: Theorems 11.1–11.2. Exact nodal inversion and the scale of low-mode errors.

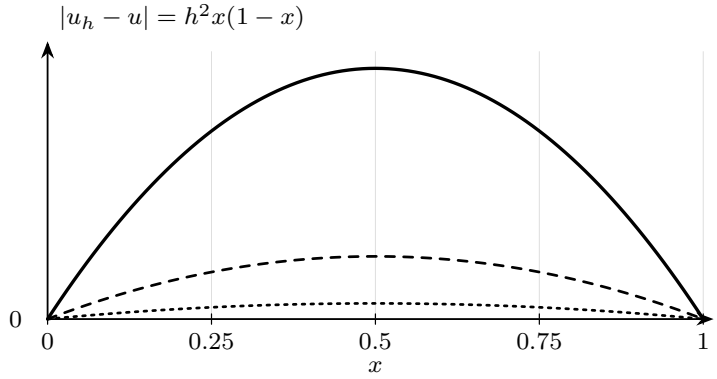
Use $u(x) = x - x^4$, which vanishes at both endpoints and solves $-u'' = 12x^2$. Its centered second difference is not exact:

$$L_h u|_{\text{nodes}} = 12x_i^2 + 2h^2.$$

Since the stencil differentiates the quadratic $x(1-x)$ exactly, the solution u_h to the sampled forcing $12x_i^2$ satisfies

$$\boxed{u_h(x_i) - u(x_i) = -h^2 x_i(1-x_i).}$$

This gives both the sign and size of the error, rather than merely a log–log slope.



grid spacing h	maximum error
$\frac{1}{4}$	$\frac{1}{64}$
$\frac{1}{8}$	$\frac{1}{256}$
$\frac{1}{16}$	$\frac{1}{1024}$

The three curves show magnitudes of the exact nodal errors for $h = 1/4, 1/8, 1/16$. Lines between grid points are visual guides, not an additional continuous numerical solution. The error is negative on every interior node.

One refinement divides this error by four

All three grids include $x = 1/2$, where the maximum absolute nodal error is $h^2/4$. It equals 1/64, 1/256, and 1/1024, respectively. Exact rational matrix inversion verifies these values independently of the closed error formula.

The manuscript also compares the discrete eigenvalues

$$\lambda_{k,h} = 4h^{-2}\sin^2(k\pi h/2)$$

with $(k\pi)^2$, with error at most $(k\pi)^4 h^2/12$. For a fixed mode k, the bound tends to zero quadratically. It does not claim a uniform small error for modes whose index grows like $1/h$.

The nodal Green identity is exact, while the discretization of the differential equation still has a truncation error. These statements concern different comparisons and do not contradict one another.

> **Read this as:** An exact inverse of the discrete operator does not make that operator equal to the differential operator. Keep algebraic inversion and continuum approximation separate.

FEYNMAN–KAC / DEPARTURE-STATE WEIGHTS

Eight paths, one weighted expectation

Main text: Theorem 12.1. A finite path sum with its time convention displayed.

Let $K_2 = \frac{1}{4}\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, start at state 0, use weights $w = (1, 1/2)$, and terminal observable $f = (2, 1)$. For three transitions, there are eight possible state sequences after the fixed initial state. We evaluate

$$\mathbb{E}_0\left[\prod_{r=0}^{2} w(X_r)\, f(X_3)\right].$$

Each path contributes its transition probability times its three departure weights times its terminal value.

Path	probability	weight	terminal	contribution
0000	27/64	1	2	27/32
0001	9/64	1	1	9/64
0010	3/64	1/2	2	3/64
0011	9/64	1/2	1	9/128
0100	3/64	1/2	2	3/64
0101	1/64	1/2	1	1/128
0110	3/64	1/4	2	3/128
0111	9/64	1/4	1	9/256

Total probability = 1; weighted terminal expectation = 311/256.

Every possible path is included. The probability column sums to one. The contribution column sums to 311/256, agreeing exactly with the transfer calculation.

Check which side of the transition gets weighted

The backward operator on observables is $D_w K_2^{\mathsf{T}}$. Therefore the expectation is the zeroth coordinate of

$$(D_w K_2^{\mathsf{T}})^3 f.$$

Its forward dual is $K_2 D_w$: weight the mass at the departing state, then move it. Weighting arrival states instead gives $D_w K_2$, a different expression. The exact tests deliberately check that the two conventions do not agree in this example.

The answer 311/256 exceeds one, but it is not a probability. The terminal observable can equal two. With $0 \le w_i \le 1$, the weighted expectation is bounded by the maximum terminal value, not necessarily by one.

This calculation certifies a finite three-step identity for a stated model. The continuous-time theorem uses the corresponding infinitesimal generator and potential, with the direction of action kept consistent.

> **Read this as:** A Feynman–Kac weight changes the value assigned to each path. Its location relative to a transition is part of the model, not a typographical choice.

SCHUR–FESHBACH / RETAINED RESOLVENT

Elimination makes a frequency-dependent term

Main text: Theorem 13.1. A two-state Schur complement with the inverse checked directly.

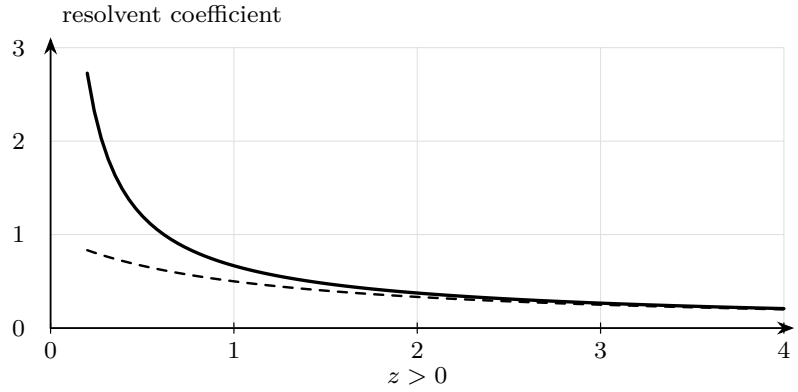
The two-state evolution from V−3 has generator

$$A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Retain the first coordinate and eliminate the second. For z away from the relevant poles, the retained entry of $(zI - A)^{-1}$ is

$$R_x(z) = \left[z + 1 - \frac{1}{z+1} \right]^{-1} = \frac{z+1}{z(z+2)}.$$

The term $\Sigma(z) = 1/(z+1)$ is the hidden coordinate's self-energy.



exact: $\dfrac{z+1}{z(z+2)}$

omit hidden state: $\dfrac{1}{z+1}$

The pole at zero survives through hidden feedback.

The full inverse gives a rational response with two poles, at 0 and −2. Its partial fractions are $R_x(z) = \frac{1}{2}z^{-1} + \frac{1}{2}(z+2)^{-1}$. The hidden intermediate inverse has its own excluded point $z = -1$.

An eliminated variable has not disappeared

If the coupling to the second coordinate were simply deleted, the retained resolvent would be $(z+1)^{-1}$. That has a different pole and a different long-time limit. The correction is frequency dependent, so it cannot be represented by one constant scalar generator on the retained coordinate.

The Schur expression first requires $z + 1 \neq 0$, as well as invertibility of the full matrix. Its simplified rational form has a removable intermediate singularity at $z = -1$, where the retained entry of the full inverse is well-defined and equals zero. Simplification does not retroactively make the eliminated block invertible there.

For $z > 0$, the Laplace transform of $(1 + e^{-2t})/2$ is exactly $R_x(z)$. The next page derives the time-domain equation producing that response.

> **Read this as:** The Schur complement is exact elimination, not independent dynamics on a smaller state space. Hidden evolution returns as a z-dependent self-energy.

EXACT RETAINED DYNAMICS / NOT A SCALAR SEMIGROUP

Hidden state returns as memory

Main text: Theorem 13.2. The retained-state memory identity, including hidden initial data.

Write the same system as

$$x' = -x + y, \qquad y' = x - y.$$

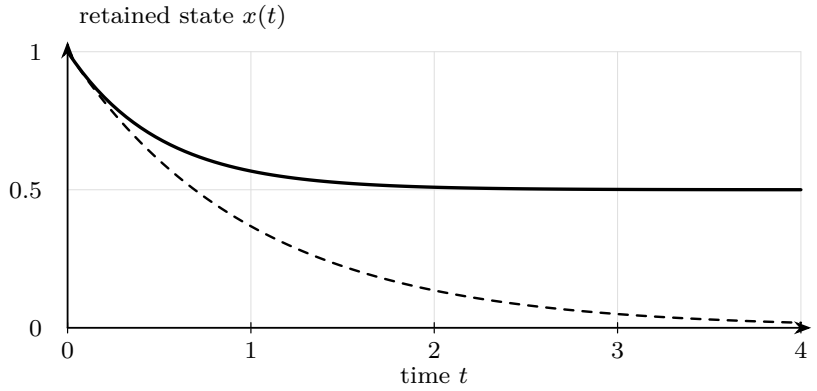
Solving the second equation first gives

$$y(t) = e^{-t}y(0) + \int_0^t e^{-(t-r)}x(r)\,dr.$$

Substitution is exact:

$$x'(t) = -x(t) + e^{-t}y(0) + \int_0^t e^{-(t-r)}x(r)\,dr.$$

The hidden initial state produces a separate forcing term. It must be retained even when the memory kernel is known.



exact: $\dfrac{1 + e^{-2t}}{2}$

naive deletion: e^{-t}

$K(t) = e^{-t}$
returns hidden excursions.

For $(x(0), y(0)) = (1, 0)$, the exact retained coordinate is $(1 + e^{-2t})/2$. The dashed curve e^{-t} solves the equation obtained by deleting the hidden coordinate rather than eliminating it.

Projection can destroy the semigroup law

Let $R(t) = (1 + e^{-2t})/2$, the response to this fixed initial preparation. At $t = \log 2$,

$$R(t) = \frac{5}{8}, \qquad R(2t) = \frac{17}{32}, \qquad R(2t) - R(t)^2 = \frac{9}{64}.$$

The full two-state propagator is a semigroup, but this retained scalar response is not. Restarting it as a scalar model silently resets the hidden coordinate, which generally carries information from the first interval.

The appendix verifies the full differential equation, initial condition, integral memory equation, and Laplace resolvent in this model. Its kernel is e^{-t}, whose transform is the self-energy on V – 21.

> **Read this as:** Exact reduction retains a record of the eliminated state through convolution and initial forcing. A memoryless approximation must justify the information it removes.

NONDIAGONAL HIDDEN DYNAMICS / POLYNOMIAL EXPONENTIALS

A repeated pole remembers a Jordan chain

Main text: Theorem 13.2 and Section 15. Functional calculus must retain nilpotent data.

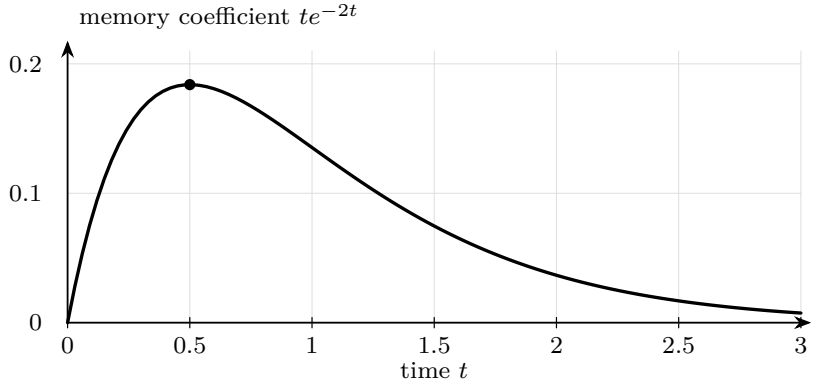
Now use a two-dimensional hidden block

$$J = -2I + N, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0.$$

Choose the entrance column $V = (0,1)^{\mathsf{T}}$ and exit row $U = (1,0)$. The hidden exponential is not $e^{-2t}I$:

$$e^{tJ} = e^{-2t}(I + tN), \qquad K_{\mathrm{mem}}(t) = Ue^{tJ}V = te^{-2t}.$$

The nilpotent link carries an input from the second hidden coordinate to the first.



$$J = -2I + N, \quad N^2 = 0$$

$$e^{tJ} = e^{-2t}(I + tN)$$

$$\widehat{K}(z) = \frac{1}{(z+2)^2}$$

A repeated pole carries a jet.

The memory kernel starts at zero, becomes positive, and peaks at $t = 1/2$. A rule that retained the eigenvalue -2 but discarded the Jordan chain would predict the zero kernel for these entrance and exit maps.

The transfer function sees pole order

The exact transform is

$$U(zI - J)^{-1}V = \frac{1}{(z+2)^2} = \int_0^\infty te^{-(z+2)t}\,dt, \qquad \operatorname{Re} z > -2.$$

The repeated pole and the polynomial factor t are the same information in two domains. For a longer nilpotent chain, higher powers of t occur with factorial denominators.

A zero value $K_{\mathrm{mem}}(0)$ does not make the entire kernel zero. Nor do stable eigenvalues alone specify every transient contribution: entrance maps, exit maps, and nilpotent structure matter. Here the full kernel is nevertheless elementary, so its peak and transform can be checked without numerical fitting.

This page is an explicit finite realization. It does not show that an arbitrary continuum memory kernel admits a finite-dimensional representation of this form.

> **Read this as:** Repeated eigenvalues may carry directional information through Jordan chains. Keep the finite jet of the functional calculus, not just its values on the spectrum.

PHASE-TYPE SURVIVAL / TWO SUCCESSIVE STAGES

A zero initial hazard can still lead to certain exit

Main text: Theorem 14.1. Survival, density, hazard, and moments have distinct meanings.

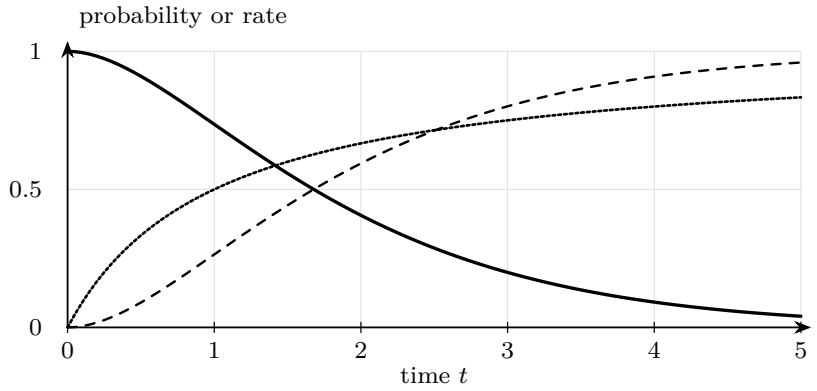
Let a transient process move through two successive rate-one stages, then leave the system. With column probabilities, the transient generator and starting vector are

$$T = \begin{pmatrix} -1 & 0 \\ 1 & -1 \end{pmatrix}, \qquad p(0) = e_0.$$

Thus $p(t) = e^{-t}(1, t)^{\mathsf{T}}$. Its total remaining mass, exit density, and hazard are

$$S(t) = (1 + t)e^{-t}, \qquad f(t) = te^{-t}, \qquad h(t) = \frac{t}{1 + t}.$$

The initial hazard is zero because an exit needs two stages, not one.



solid: survival $S(t)$

dashed: absorbed mass

dotted: hazard $h(t)$

$h(0) = 0$, but eventual exit has probability one.

Survival falls from one to zero, while absorbed mass $1 - S$ rises to one. The dotted hazard is an instantaneous rate, not a probability, even though it shares the displayed numerical scale. Time is measured in inverse stage-rate units.

Missing transient mass is absorbed mass

The column sums of T are 0 and -1. The exit-rate row is therefore $(0, 1)$, and $f(t) = (0, 1)p(t)$. Adding an absorbing state restores a three-state probability vector with total mass one.

Exact integration gives

$$\int_0^\infty f(t)\, dt = 1, \qquad \mathbb{E}[\tau] = 2, \qquad \mathbb{E}[\tau^2] = 6, \qquad \mathrm{Var}(\tau) = 2.$$

Equivalently, the mean is $\mathbf{1}^{\mathsf{T}}(-T)^{-1}e_0$. A single exponential waiting time with mean two would have constant hazard $1/2$ and variance four, so matching the mean alone would not reproduce this distribution.

The hazard formula is used where $S(t) > 0$. The model has that property for every finite $t \geq 0$. No division by an already vanished survival mass is required.

> **Read this as:** Survival tracks mass, density tracks exits per unit time, and hazard conditions on survival. Equal means or equal initial rates do not identify a waiting-time law.

HOLOMORPHIC CALCULUS / A RECTANGULAR COMPARISON

Naturality also transports derivatives

Main text: Theorem 15.1. Exact intertwining passes to the specified functional calculus.

Consider a fine operator, a coarse operator, and a rectangular map

$$A_f = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 9 \end{pmatrix}, \qquad A_c = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \qquad C_r = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Direct multiplication gives $C_r A_f = A_c C_r$. For $F(z) = 1/(1+z)$, which is defined at both spectra, compute the matrices by inversion:

$$F(A_f) = \begin{pmatrix} 1/3 & -1/9 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/10 \end{pmatrix}.$$

A_f

2	1	0
0	2	0
0	0	9

$F(A_f)$ for $F(z) = (1+z)^{-1}$

$\frac{1}{3}$	$-\frac{1}{9}$	0
0	$\frac{1}{3}$	0
0	0	$\frac{1}{10}$

$$F(2) = 1/3$$

$$F'(2) = -1/9$$

The dashed block is $F(A_c)$, not just $F(2)I$.

The discarded eigenvalue 9 contributes $1/10$ only on the fine side. On the retained Jordan block, the off-diagonal entry is $F'(2) = -1/9$, not zero. The rectangular comparison preserves that entry.

Why spectral values are not enough

On the retained block $A_c = 2I + N$, with $N^2 = 0$, one has

$$F(A_c) = F(2)I + F'(2)N = \tfrac{1}{3}I - \tfrac{1}{9}N.$$

Hence $C_r F(A_f) = F(A_c) C_r$. Replacing the block by $F(2)I$ would lose the nonzero derivative term and no longer give the actual inverse of $I + A_c$.

For polynomials, naturality follows by transporting every operator power. The manuscript's finite holomorphic statement uses finite Taylor data at each primary block and the corresponding nilpotent orders. This example verifies a rational instance with an actual inverse rather than asserting that arbitrary scalar substitutions are meaningful.

The same mechanism explains why the Jordan memory kernel on V – 23 contains te^{-2t}. Exponentiation also has nonzero derivatives at repeated spectral values.

> **Read this as:** A functional calculus acts on operators, including their nilpotent parts. Exact intertwining transports that full action, not merely a list of scalar eigenvalue images.

END-TO-END MODEL / TWO OBSERVATIONS

One complete coarse filter

Main text: Proposition 9.3 with Theorems 5.1 and 8.2. Markov closure and likelihood sufficiency are separate.

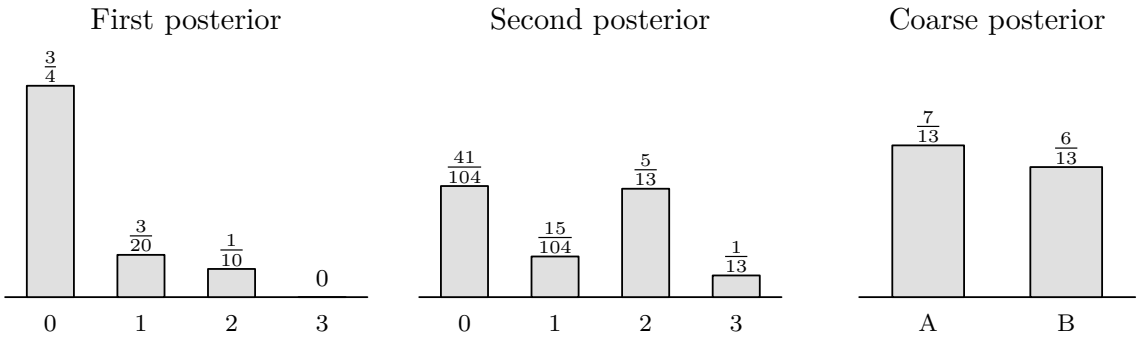
Use the four-state kernel K and aggregation C from V – 5, starting at $p_0 = e_0$. Observe two likelihood vectors constant inside both packets:

$$\lambda^{(1)} = (3/4, 3/4, 1/4, 1/4), \qquad \lambda^{(2)} = (1/2, 1/2, 1, 1).$$

Predict, then update at each observation. The fine calculation gives

$$p_1 = \tfrac{1}{20}(15, 3, 2, 0)^{\mathsf{T}}, \qquad p_2 = \tfrac{1}{104}(41, 15, 40, 8)^{\mathsf{T}}.$$

Evidence $Z_1 = 5/8$, $Z_2 = 13/20$, and $Z_1 Z_2 = 13/32$



Within-packet probabilities change during the two observations, but the packet totals agree exactly with an independent two-state filter. The second coarse posterior is $(7/13, 6/13)$.

Check the normalizers, not only the final vector

The first and second predictive evidences are

$$Z_1 = \frac{5}{8}, \qquad Z_2 = \frac{13}{20}, \qquad Z_1 Z_2 = \frac{13}{32}.$$

The unnormalized product $D_2 K D_1 K p_0$ has mass $13/32$ and normalizes to p_2. Because the likelihood entries here lie in $[0, 1]$, they can serve as probabilities of the specified observations. This makes the product their joint evidence in the stated model.

Two distinct equalities justify the coarse computation. First, $CK = K_c C$ transports prediction. Second, the likelihood is constant on each packet, so $CD_r = \overline{D}_r C$ transports observation weighting. Their composition preserves both the posterior packet totals and the evidence factors.

The coarse filter does not recover the within-packet shape. If a later observation distinguishes states inside a packet, those missing probabilities can become necessary, as the counterexample on V – 10 shows.

> **Read this as:** Exact filtering at a coarse level needs both dynamical lumpability and observation sufficiency. Agreement under one step does not excuse testing the other.

REFINEMENT LIMITS / WHAT THE FINITE MODELS DO NOT PROVE

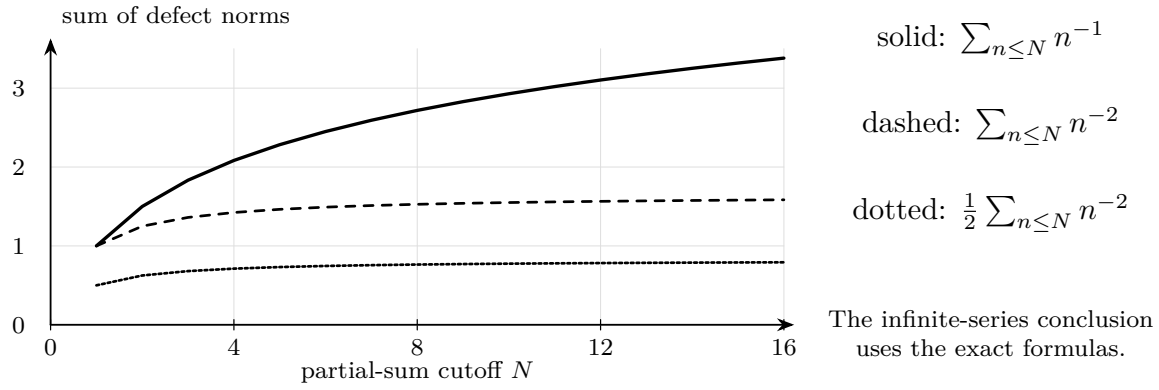
Small at each level is not summable across levels

Main text: Corollary 10.5 and Section 17. Summability and uniform control are additional data.

For each $n \geq 1$, compare the two scalar operators $A_n = 0$ and $B_n = 1/n^2$ through the identity. Their ordinary defects have sizes $1/n^2$, while their square-root defects have sizes $1/n$:

$$\sum_n |A_n - B_n| < \infty, \qquad \sum_n |A_n^{1/2} - B_n^{1/2}| = \infty.$$

These are independent scalar comparisons, not a constructed inverse system. They isolate an error-scale obstruction without claiming a coherent refinement tower.



Partial sums through $n = 16$ illustrate the contrast. The convergence and divergence statements follow from the displayed series, not from the finite plot. A common positive gap gives the third, summable upper bound.

A uniform hypothesis can change the conclusion

Replace the pair by $A_n = 1$ and $B_n = 1 + 1/n^2$. Then

$$\left|B_n^{1/2} - A_n^{1/2}\right| = \frac{1/n^2}{\sqrt{1 + 1/n^2} + 1} \leq \frac{1}{2n^2}.$$

This is the elementary form of the uniform-gap mechanism on V – 16. Without such information, transferring a summable generator-defect budget through a fractional function can fail.

The formal release keeps three original research questions separate: realization of broader transfer semigroups, uniform stochastic-refinement control, and latent packet closure. The exact finite examples here do not settle their existence, domain, topology, or uniformity requirements.

Likewise, exact hidden-state elimination only yields the finite memory models that were supplied. It does not establish a finite latent realization for every memory law. A complete list of matrix identities can still leave an infinite-dimensional approximation problem untouched.

> **Read this as:** A finite example can expose a missing hypothesis. It cannot supply a projective-limit construction, a uniform constant, or a continuum realization merely by passing at several levels.

READER'S GUIDE / CALCULATION, DRAWING, AND FORMAL CONTRACT

How to use the evidence

Source release: Paper VI v0.5.0, Checked Fifth Block. The original manuscript and proof boundaries are preserved.

This appendix is an explanatory layer. The existing 34-page manuscript remains unchanged in the combined edition, including its figures and formal-status appendix. Source lines beneath these titles link back to the relevant manuscript pages in that edition.

The chosen release records all 33 scoped results complete, with 341 supporting theorem declarations in 47 proof modules and a saved audit of 782 namespace declarations. Those are inherited verification records, not a new Lean run and not counts of historically novel results. The three original conjectures retain their stated status.

Read each kind of evidence at its actual strength

Evidence	What is checked here	What it does not establish
Exact matrices	Stochasticity, intertwining, projector identities, inverse formulas, and specific defects.	The same properties for arbitrary partitions or unbounded operators.
Rational algebra	Posterior/evidence identities, polynomial integrals, nodal errors, and complete finite path sums.	An unstated observation model or a continuum convergence theorem.
Symbolic formulas	The displayed exponentials, Laplace transforms, derivatives, and finite spectral powers under stated conditions.	A general-purpose formal proof of every analytic identity.
Vector drawings	Reproducible geometry and sampled curves from the explicit models.	Numerical quadrature certificates or an exhaustive infinite-time experiment.
Saved Lean records	The original contract, source files, dependency pins, and earlier verification logs.	Fresh kernel verification of this editorial addition.

Reproduce a calculation before adapting it

The visual-appendix tree contains the mathematical models, drawing generator, and exact-example verifier. Its README records their paths and execution commands. The verifier currently checks 204 exact or symbolic assertions and rejects 11 deliberately false controls. Its JSON record names every check and its finite scope.

The finite families are explicit: three-step Feynman–Kac paths are fully enumerated, selected rational priors are checked exhaustively within their stated denominator bound, and Green inverses are checked on four named grids. No finite count is used as a substitute for a general theorem.

To change a model, specify the state convention, partition, likelihood meaning, topology, and domain first. Then rerun the calculations and inspect the regenerated drawings. A preserved formula with changed assumptions is not automatically a preserved result.

> **Read this as:** Use the appendix to see why a hypothesis matters and to compute a small case. Use the manuscript and its formal map for the scope of the general claim.